For both configurations generated from the 19 EEG channels, p-values greater than the significance threshold of 0.05 were obtained. This means that no statistically significant differences were found between the models. Although the networks differ in their metric values, the differences are not large enough to be considered statistically significant.

The same situation occurred for the 4-channel eyes-open data. The p-value was 0.8654, which significantly exceeds the significance threshold. Such a high value indicates that the differences between the models were very minor. None of the architectures showed any advantage over the others.

For the 4-channel EC configuration, the p-value obtained in the Friedman test was 0.0269, which was lower than the established significance threshold. This result indicates that there are statistically significant differences between at least two models.

Table 1. Friedman test results

Configuration	p-value
19 channels EO	0,2938
19 channels EC	0,3782
4 channels EO	0,8654
4 channels EC	0,0269

For the statistically significant configuration, an additional multiple comparisons test was performed with Bonferroni correction. The correction reduces the risk of obtaining false positives resulting from performing multiple comparisons simultaneously.

The post-hoc analysis revealed only one pair of models between which a statistically significant difference was found. For the VGG16 and InceptionV3 networks, the p-value was 0.0179. For the remaining pairs, the p-value was higher than the significance threshold, meaning there is no basis to claim significant differences between the models.

The VGG16 model for the 4-channel EC data achieved an accuracy of 74.08%, whereas the InceptionV3 network achieved 52.25%. The Friedman and multiple comparisons tests indicated that the difference between these models was statistically significant. The lack of significant differences among the remaining models means there is no basis to state that the differences in classification accuracy stem from the superiority of one model over another.

Table 2. Results of the multiple comparisons test (1=4-layers CNN, 2=10-layers CNN, 3=ResNet50V2, 4=VGG16, 5=InceptionV3)

Model A	Model B	Lower confidence limit	Difference	Upper confidence limit	p-value
1	2	-0.9016	-0.2013	0.4990	1.0000
1	3	-0.6418	0.0584	0.7587	1.0000
1	4	-1.1159	-0.4156	0.2847	0.9574
1	5	-0.3366	0.3636	1.0639	1.0000
2	3	-0.4405	0.2597	0.9600	1.0000
2	4	-0.9146	-0.2143	0.4860	1.0000
2	5	-0.1353	0.5649	1.2653	0.2354
3	4	-1.1743	-0.4740	0.2263	0.5742
3	5	-0.3951	0.3052	1.0055	1.0000
4	5	0.0789	0.7792	1.4795	0.0179

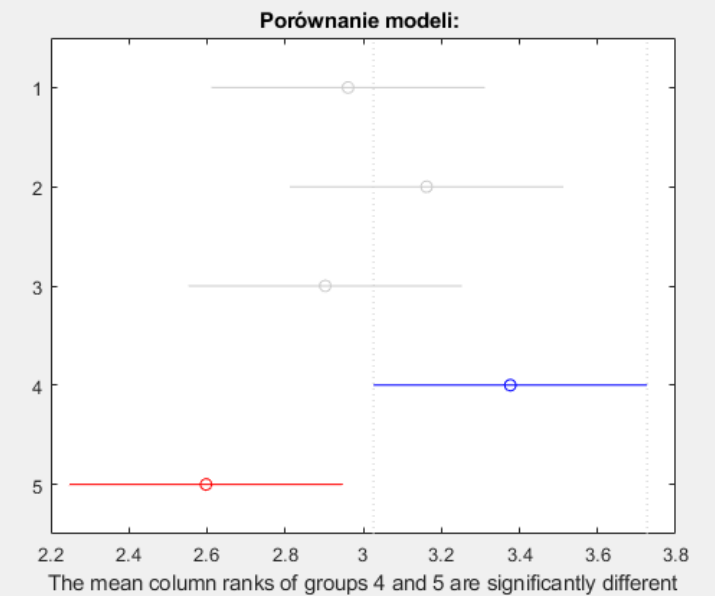


Fig. 1. Graphical representation of the multiple comparisons test results. Models showing a statistically significant difference are coloured.